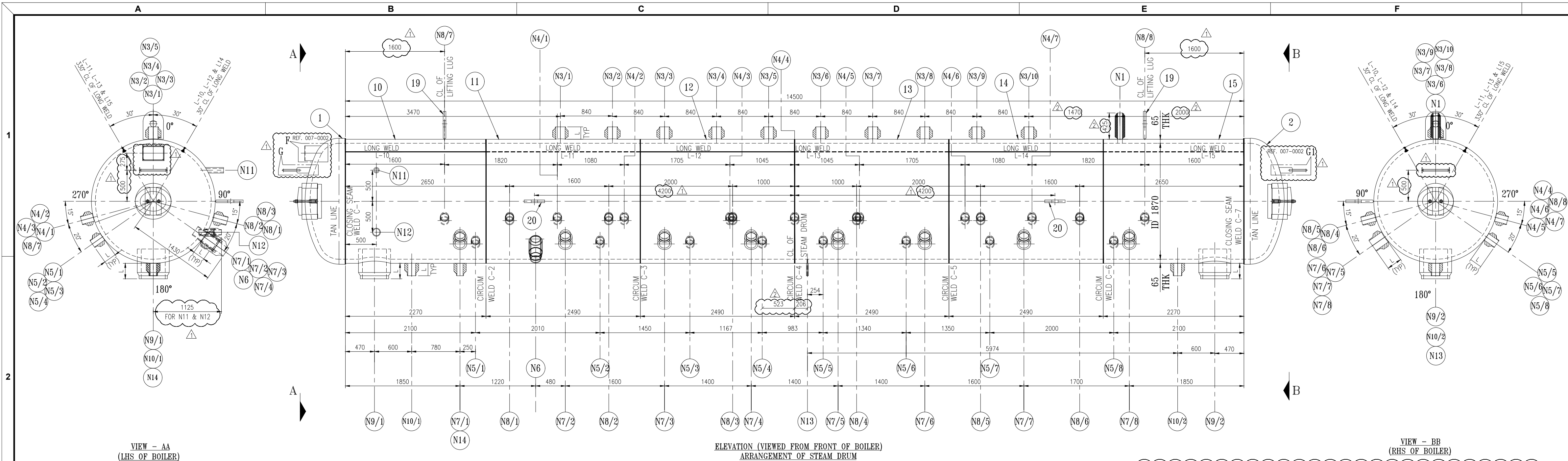
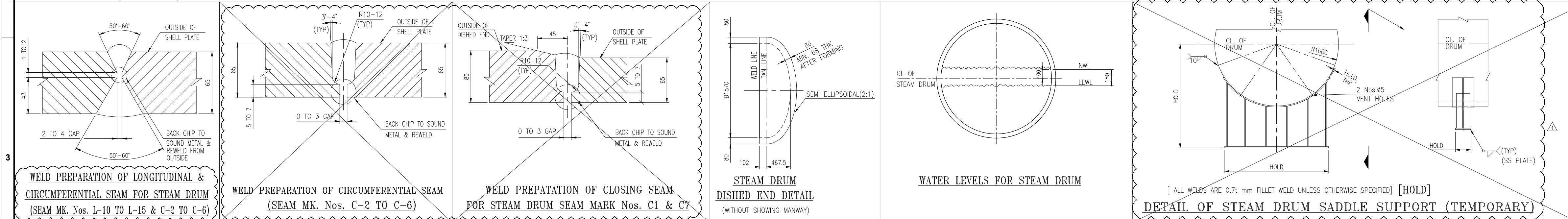




REF. DOCUMENT				
DOCUMENT	TITLE			REV.
NOTES:				
01. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.				
02. FOR GENERAL NOTES & MATERIAL SPECIFICATIONS REF. DOCUMENT ID : JB1174-50010100-007-0002				
03. MAXIMUM ALLOWABLE WORKING PRESSURE INDICATED IN DESIGN DATA IS EXCLUSIVE OF STATIC HEAD.				
04. THE SUPPORT ARRANGEMENT WILL NOT RESULT IN UNDUE STRESS BEING IMPOSED ON THE DRUM.				
05. FOR GENERAL WELDING, INSPECTION & TESTING, TOLERANCE, PAINTING AND OTHER NOTES REFER DWG NO. JB1174-50010100-007-0002.				
06. AS WE CONSIDER THE ELEVATION OF STEAM DRUM IN THE REFERENCE OF OTHER EQUIPMENT, HENCE STATIC PRESSURE OF THE STEAM DRUM CONSIDERED AS ZERO.				
07. PROVIDE WOODEN BLOCKS BELOW SADDLE DURING TRANSPORTATION & HANDLING AT SHOP & SITE SUCH THAT IT WILL NOT DAMAGE DOWNCOMER NOZZLE. DOWNCOMER NOZZLE MUST BE COME INSIDE SADDLE + WOODEN BLOCK HEIGHT.				
08. REQUIREMENT FOR NON DESTRUCTIVE EXAMINATION OF DRUM ATTACHMENT WELD & OTHER WELDS, REFER HZI ITP DOCUMENT QPW-HZI-50103514-D-0				
09. LARGE OPENINGS OF ALL NOZZLES EXCEPT MARK NOS. N11, N13 & N14 ARE INHERENTLY COMPENSATED. THOSE NOZZLE ARE FULL PENETRATION WELD JOINT & FULL PENETRATION FROM ROOT SIDE SHALL BE ENSURED DURING WELDING AND SHALL BE GROUND FROM INSIDE. ROOT RUN SHALL BE BY TIG PROCESS AND SHALL BE DP TESTED BEFORE FULL WELDING. AFTER WELDING, END CORNERS SHALL BE SMOOTH GROUND WITH 3 mm RADIUS.				
MATERIAL SPECIFICATIONS :				
DESCRIPTION	SIZE	SPECIFICATION	MIN. CALCULATED THK.	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	62.342 mm	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	62.342 mm	FOR STEAM DRUM



AS PER FIG No. : PW16.1 (a)		AS PER FIG No. : PW16.1 (a)		AS PER FIG No. : PW16.1 (a)		AS PER FIG No. : PW16.1 (a)	
W/S-N3/1, W/S-N3/2, W/S-N3/3, W/S-N3/4, W/S-N3/5, W/S-N3/6, W/S-N3/7, W/S-N3/8, W/S-N3/9, W/S-N3/10, W/S-N4/1, W/S-N4/2, W/S-N4/3, W/S-N4/4, W/S-N4/5, W/S-N4/6, W/S-N4/7, W/S-N4/8, W/S-N4/9, W/S-N4/10, W/S-N5/1, W/S-N5/2, W/S-N5/3, W/S-N5/4, W/S-N5/5, W/S-N5/6, W/S-N5/7, W/S-N5/8, W/S-N5/9, W/S-N5/10, W/S-N6/1, W/S-N6/2, W/S-N6/3, W/S-N6/4, W/S-N6/5, W/S-N6/6, W/S-N6/7, W/S-N6/8, W/S-N6/9, W/S-N6/10, W/S-N7/1, W/S-N7/2, W/S-N7/3, W/S-N7/4, W/S-N7/5, W/S-N7/6, W/S-N7/7, W/S-N7/8, W/S-N7/9, W/S-N7/10, W/S-N8/1, W/S-N8/2, W/S-N8/3, W/S-N8/4, W/S-N8/5, W/S-N8/6, W/S-N8/7, W/S-N8/8, W/S-N8/9, W/S-N8/10, W/S-N9/1, W/S-N9/2, W/S-N9/3, W/S-N9/4, W/S-N9/5, W/S-N9/6, W/S-N9/7, W/S-N9/8, W/S-N9/9, W/S-N9/10, W/S-N10/1, W/S-N10/2, W/S-N10/3, W/S-N10/4, W/S-N10/5, W/S-N10/6, W/S-N10/7, W/S-N10/8, W/S-N10/9, W/S-N10/10, W/S-N11, W/S-N12, W/S-N13, W/S-N14		W/S-N3/1 & W/S-N3/2		W/S-N3/1 & W/S-N3/2		W/S-N3/1 & W/S-N3/2	
WELDING DETAIL OF NOZZLE TO DRUM [FOR MARK No. N1] [FOR WELD No.- REFER TABLE-2]		WELDING DETAIL OF NOZZLE TO DRUM [FOR MARK Nos. N3/1 TO N3/10, N4/1 TO N4/7, N5/1 TO N5/8, N7/1 TO N7/8, N8/1 TO N8/8, N10/1, N10/2, N12 & N14] [FOR WELD No.- REFER TABLE-2]		WELDING DETAIL OF NOZZLE TO DRUM [FOR MARK Nos. N9/1, N9/2] [FOR WELD No.- REFER TABLE-2]		WELDING DETAIL OF NOZZLE TO DRUM [FOR MARK No. N6 & N7/1 TO N7/8] [FOR WELD No.- REFER TABLE-2]	

TABLE - 1																							
S.No.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (D) (mm)	Nozzle Thickness (t) (mm)	Nozzle Size (ID) (D) (mm)	Schedule of Nozzle	Connecting Pipe Size (mm)	Connecting Pipe Thickness (mm)	Connecting Pipe ID (mm)	Projection from C/End of Drum (U) (mm)	Drum Vertical Axis (X) (mm)	Drum Horizontal Axis (Y) (mm)	Calculated Length (R) (mm)	Finished Length of the Nozzle (M) (mm)	MFG Bone for Projection (E) (mm)	Weld Dim. (t1) (mm)	Weld Dim. (t2) (mm)	Weld Dim. (t3) (mm)	Projection - In (W) (mm)	Projection - Out (P) (mm)	
1	N1	SAFETY VALVE NOZZLE	STEAM DRUM SHELL	1	160.00	41.90	76.2		127.0	25.40	76.2	425			425	425					14		
2	N3/1 TO N3/10	SATURATED STEAM STUB	STEAM DRUM SHELL	10	246.36	50.0	146.36	-	168.3	10.97	146.36	200	-	-	230	200	-	-	-	-	14	0	
3	N4/2,N4/3,N4/5 & N4/6	3rd PASS REAR WALL TOP RISER	STEAM DRUM SHELL	4	222.24	50.0	122.24	-	141.3	9.53	122.24	200	-	-	230	200	-	-	-	-	14	0	
4	N4/1,N4/4 & N4/7	4th PASS LEFT WALL, RIGHT TOP RISER	STEAM DRUM SHELL	3	222.24	50.0	122.24	-	141.3	9.53	122.24	200	-	-	230	200	-	-	-	-	14	0	
5	N5/2 TO N5/7	2nd PASS REAR WALL ROOF RISER	STEAM DRUM SHELL	6	222.24	50.0	122.24	-	141.3	9.53	122.24	200	-	-	230	200	-	-	-	-	14	0	
6	N5/1 & N5/8	3rd PASS LEFT & RIGHT SIDE WALL TOP RISER	STEAM DRUM SHELL	2	222.24	50.0	122.24	-	141.3	9.53	122.24	200	-	-	230	200	-	-	-	-	14	0	
7	N6	FEED INLET NOZZLE WITH THERMAL SLEEVE	STEAM DRUM SHELL	1	299	55	189	-	168.3	10.97	146.36	430			REF. DRG. No.009-0001							125	

LOADING AS PER PG-22

PLATE FOR TEST COUPON (PTC)

WEIGHT OF DRUM

WATER HOLDING CAPACITY

GOVERNING LIGAMENT EFFICIENCY

NIL

NIL EXCEPT MAXIMUM ALLOWABLE WORKING PRESSURE AND HYDROSTATIC HEAD

NOT REQUIRED AS PER PW-53.1

56770.27 kgs (EMPTY), 98874.43 kgs (FULL)

41.52 m³ (APPROX)

1

OUTSIDE OF DISHD END

45

50°-60°

TAPER 1:3

65

10.2

43

2 TO 4 GAP

BACK CHIP TO SOUND METAL & REWELD FROM OUTSIDE

50°-60°

80

WELD PREPARATION OF CLOSING SEAM FOR STEAM DRUM FOR MARK Nos. C1 & C7

ASME SECTION I, CLAUSE PG-22 : EXTERNAL LOADING :-

BOILER IS SUBJECTED TO INTERNAL PRESSURE AND HYDROSTATIC HEAD ONLY. NO OTHER EXTERNAL LOAD IS ACTING ON THE BOILER. LOADING DUE TO PIPING IS NOT SIGNIFICANT AS THE PIPING STRESS ANALYSIS WILL BE DONE SEPARATELY AND FLEXIBILITY WILL BE MAINTAINED.

LIST OF REFERENCE DRAWINGS

DRAWING DOCUMENT ID	DESCRIPTION
JB1174-50010100-000-0001	ARRANGEMENT & DETAIL OF MANIFOLD ON DRUMS
JB1174-50010100-007-0002	MATERIAL LIST OF STEAM DRUM
JB1174-50010100-009-0001	DETAIL OF STEAM DRUM NOZZLES

6

8	N7/1 TO N7/8	1st PASS FRONT WALL ROOF RISER	STEAM DRUM SHELL	8	303.7	55.0	193.7	—	219.1	12.7	193.7	215	—	—	245	215	—	—	—	—	21	0	△	
9	N8/1 TO N8/6	1st & 2nd PASS SIDE WALL RISER	STEAM DRUM SHELL	6	222.24	50.0	122.24	—	141.3	9.53	122.24	200	—	—	230	200	—	—	—	—	14	0		
10	N8/7 & N8/8	1st & 2nd PASS SIDE WALL RISER	STEAM DRUM SHELL	2	222.24	50.0	122.24	—	141.3	9.53	122.24	200	—	—	230	200	—	—	—	—	14	0		
11	N9/1 & N9/2	DOWN COMER NOZZLE	STEAM DRUM SHELL	2	517.14	65	387.14	—	457.0	34.93	387.14	250	—	—	530	480	—	—	—	—	45	165	△	
12	N10/1 & N10/2	DOWN COMER NOZZLE (EVAPORATOR)	STEAM DRUM SHELL	2	222.24	50.0	122.24	—	141.3	9.53	122.24	200	—	—	230	200	—	—	—	—	14	0		
13	N11	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHEEL	1	88.9	15.24	58.42	SCH.XXS	88.9	15.24	58.42	1125	—	—	500	315	286	—	—	—	11	0	△	
14	N12	LEVEL TRANSMITTER (WATER) WITH THERMAL SLEEVE	STEAM DRUM SHEEL	1	200	45.53	108.9	—	88.9	15.24	58.42	1125	REF. DRG. No.009-0001										25	△
15	N13	CBD NOZZLE	STEAM DRUM SHEEL	1	33.4	9.09	15.22	SCH.XXS	33.4	4.55	24.3	200	—	—	230	200	—	—	—	—	7	0		
16	N14	EMERGENCY DRAIN	STEAM DRUM SHELL	1	88.9	11.13	66.64	SCH.160	88.9	7.62	73.66	200	—	—	230	200	—	—	—	—	8	0	△	

TABLE - 2

NOZZLE SHOP WELD NO.	
NOZZLE NO.	WELD NO.
N1	W/S-N1
N3/1	W/S-N3/1
N3/2	W/S-N3/2
N3/3	W/S-N3/3
N3/4	W/S-N3/4
N3/5	W/S-N3/5
N3/6	W/S-N3/6
N3/7	W/S-N3/7
N3/8	W/S-N3/8
N3/9	W/S-N3/9
N3/10	W/S-N3/10
N4/1	W/S-N4/1
N4/2	W/S-N4/2
N4/3	W/S-N4/3
N4/4	W/S-N4/4
N4/5	W/S-N4/5
N4/6	W/S-N4/6
N4/7	W/S-N4/7
N5/1	W/S-N5/1
N5/2	W/S-N5/2
N5/3	W/S-N5/3
N5/4	W/S-N5/4
N5/5	W/S-N5/5
N5/6	W/S-N5/6
N5/7	W/S-N5/7
N5/8	W/S-N5/8

NOZZLE SHOP WELD NO.	
NOZZLE NO.	WELD NO.
N6	W/S-N6
N7/1	W/S-N7/1
N7/2	W/S-N7/2
N7/3	W/S-N7/3
N7/4	W/S-N7/4
N7/5	W/S-N7/5
N7/6	W/S-N7/6
N7/7	W/S-N7/7
N7/8	W/S-N7/8
N8/1	W/S-N8/1
N8/2	W/S-N8/2
N8/3	W/S-N8/3
N8/4	W/S-N8/4
N8/5	W/S-N8/5
N8/6	W/S-N8/6
N8/7	W/S-N8/7
N8/8	W/S-N8/8
N9/1	W/S-N9/1
N9/2	W/S-N9/2
N10/1	W/S-N10/1
N10/2	W/S-N10/2
N11	W/S-N11
N12	W/S-N12
N13	W/S-N13
N14	W/S-N14